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Consequences of Social Frontiers Segregation for Labour Income: Evidence From Oslo

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ABSTRACT

We study the impact on earned income of living in neighbourhoods characterised by a special type of residential segregation, called social frontiers. Possible labour market outcomes of living in neighbourhoods marked by social frontiers have not been previously studied. Social frontiers represent abrupt transitions between adjacent neighbourhoods and arise when there are significant and substantial spatial divisions or 'cliff-edges' in the proportion of population groups in the contiguous neighbourhoods. This type of segregation, where highly different demographic groups are located close to each other, could cause reduced inter-group contact, which may impact labour market attainment. We use rich individual-level administrative data from Oslo (Norway) and advanced statistical methods for systematically identifying social frontiers along immigrant-native lines. Our causal design applies propensity score matching (PSM) to control for self-selection into neighbourhoods, which might otherwise bias the results. We find that residing on the immigrant-dense side of a social frontier has a statistically significant negative effect on the earnings of men, but overall, the effects are small.

1 | Introduction

Income inequality has increased in many Western countries, and research shows that variation in wage income has contributed to greater income inequality (André et al. 2018). In Norway, where income inequality has been low for long periods of time, more recent data show clear tendencies towards increased income inequality. This development is partly related to a growing number of migrants and refugees in Norway (Statistics Norway SSB 2018). A large proportion of these groups have earnings tilting towards the lowest end of the income distribution, and compared to natives, there is a consistent gap in labour market participation rates. According to Statistics Norway SSB (2018), the trajectories of income inequality depend on immigrants' success in the labour market (Battisti et al. 2022).

In this paper, we focus on income inequality by studying the impacts on work income of urban residential segregation statistically defined as social frontiers (SF), which puts an emphasis on socioeconomic processes taking place in and between residential neighbourhoods. This research has strong relevance since many migrants settle in immigrant-dense neighbourhoods of cities, augmenting already existing residential segregation patterns. This development runs the risk of hampering social exchange between natives and immigrant groups even more, which in turn may slow down labour market integration (Skifter-Andersen et al. 2016; Andersson et al. 2014).

Although there are numerous studies focusing on employment outcomes of living in ethnically segregated neighbourhoods (Goel and Lang 2019). To our knowledge, this is the first study investigating the causal effects on earnings of living in neighbourhoods close to SF (Dean et al. 2019). SF represent a type of residential segregation, where abrupt differences in neighbourhood composition appear in the urban landscape. Adjacent neighbourhoods may differ in terms of social,

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economic, cultural or ethnic characteristics, as well as in the specific combinations arising from the intersections of these features (Dean et al. 2019). Here, we study differences in terms of ethnic composition operationalised by information on country of origin.

The refinement of measures able to identify and describe different segregation patterns is a central issue in the literature (Ellen et al. 2016). However, SF cannot be identified via indexes (e.g., the dissimilarity index) commonly used in segregation research. This is mainly due to their aspatial nature: very different spatial patterns of segregation could exist in two cities (e.g., presence or absence of SF), even if they have the same value of the aggregate dissimilarity index (Dean et al. 2019). Our two-step method of locating SF through the use of a local spatial statistics estimator is superior to those used in previous studies (Zabel 2008).

Little is known about how SF influence work income. However, the few empirical studies existing show adverse effects on mental health (Maguire et al. 2016), antisocial behaviour (Legewie and Schaeffer 2016) and crime (Dean et al. 2019). Moreover, impacts on patterns of household residential mobility (Olner et al. 2023), housing prices (Myatt et al. 2025) and the propensity to commute (Severinsen et al. 2025) of SF vary depending on SF existence and frontier side residence.

Selective migration (Skifter Andersen 2019) and sorting (Kuminoff et al. 2013) are central driving forces for segregation. Individuals, both natives and immigrants, having different levels of work income, motivations, skills and peer preferences, self-select into neighbourhoods. Thus, this self-selection makes inferences biased if we were to simply compare means. We use propensity score matching (PSM) methodology and rich individual-level longitudinal administrative data from Statistics Norway to control for self-selection into immigrant-dense neighbourhoods.

Our findings suggest that the consequences for labour market outcomes vary between immigrants and natives and along gender lines. It is more constraining on income for native men to live on the immigrant-dense side of an SF neighbourhood, whereas no constraining outcome is observed for women or immigrant men.

2 | Theoretical Framework

Residential segregation arises when population groups are physically separated from one another (Skifter Andersen 2019), and such separation often entails limited intergroup interaction (White 1983). Evidence from Europe, and Scandinavia in particular, shows that local residential context matters for a wide range of economic outcomes (Edin et al. 2003; Musterd et al. 2008; van Ham and Manley 2010; G. C. Galster 2012; Andersson et al. 2014; Zuccotti and Platt 2017; Andersson et al. 2019; Nordvik et al. 2019; Battisti et al. 2022). While traditional measures of segregation (e.g., measures of ethnic concentration or the index of dissimilarity) capture broad patterns of demographic composition, they do not focus on spatial proximity of neighbourhoods, and may, hence, obscure abrupt transitions that exist at very local spatial scales (Reardon and O'Sullivan 2004).

We extend this literature by focusing on (SF), a distinct form of segregation defined as sharp, statistically significant demographic discontinuities between adjacent neighbourhoods (Dean et al. 2019; Iyer and Pryce 2024). These cliff-edge boundaries may be shaped by historical planning decisions or by natural and physical features such as roads, rivers or railway tracks (e.g., Ananat 2011), and they separate contrasting population groups located in contiguous neighbourhoods. SF highlight abrupt spatial transitions. Such boundaries may restrict everyday contact and create distinct opportunity structures for residents living in core areas compared with those living near the frontier.

A well-known example of SF is the historical division between Catholics and Protestants in Belfast, where sectarian social tensions predate the construction of the physical 'peace walls' (Olner et al. 2023). The Belfast case illustrates the frontier intuition: even in the absence of formal barriers, residents learn and reproduce territorial demarcations that define 'ours' and 'theirs', calibrate their daily routines to avoid crossing into the other group's area, and experience heightened vigilance in interface zones. These lived boundaries limit intergroup contact and opportunities for social cohesion, and they channel access to services, schools and job information along within-group lines. When physical walls later formalise these social boundaries, the underlying social distance and the asymmetric control that one side may exercise over spaces and symbols have already shaped networks, trust and mobility (e.g., Shirlow and Murtagh 2006). This intuition helps explain why very small geographic distances can translate into large social distances, and why the effects of frontiers are poorly captured by aggregate segregation indices or measures of immigrant composition in the neighbourhood of residence.

Empirically, frontier dynamics have observable behavioural correlates and outcomes. In Rotterdam, individuals living on the immigrant-dense side of a frontier were significantly less likely to move than those living on the native-dense side, thereby increasing their cumulative exposure to local neighbourhood effects (Olner et al. 2023). Severinsen et al. (2025) argue that labour market accessibility is likely to be similar on both sides of the frontier due to geographic proximity. Consequently, differences in commuting behaviour and opportunity structures are assumed to be driven by social mechanisms rather than by physical accessibility. In general, neighbourhoods benefit from high spatial labour market mobility, inter alia, because commuters share information about the wider labour market within their local neighbourhoods, which may improve the job-matching process. Descriptive statistics reveal that workers in immigrant-dense frontier areas commute shorter distances on average, suggesting that they rely more on localised, co-ethnic job networks. By contrast, workers on the native-dense side tend to engage in broader, more individualistic job-search strategies, which may influence labour market attainment. These results are particularly pronounced for non-Western immigrant women and local native men, underscoring the interaction between gender, ethnicity and spatial mobility.

To understand the impact of SF on labour market outcomes, we draw on Iyer and Pryce's (2024) theoretical framework, which outlines propositions concerning the social, economic, psychological and life-course consequences of SF. We focus on mechanisms related to reduced intergroup contact, asymmetrical territoriality and heightened awareness of inequality. Social networks and intergroup contact are important, for example, because a large proportion of workers obtain jobs

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through both formal and informal networks and referrals (Dustmann et al. 2016).

Reduced opportunities for positive intergroup contact are central to this framework. According to Intergroup Contact Theory (Allport 1954), positive interactions reduce prejudice and build trust; however, sharp spatial boundaries hinder such encounters and may instead foster territorial avoidance (Sampson 2012). SF also weaken indirect forms of contact, such as extended and vicarious contact, which are known to improve intergroup attitudes even in the absence of direct interaction (Vezzali et al. 2015). Because frontier zones sharply divide population groups, few residents occupy positions that enable cross-boundary bridging. While residents living near the frontier might, in principle, act as bridge-builders, the intensity of the boundary reduces both their presence and their effectiveness (Dean et al. 2019).

Limited contact constrains access to majority-group social capital, which is central to job referrals and informal information flows (Granovetter 1973). As a result, individuals living in SF neighbourhoods are more likely to depend on within-group networks for social and economic support. This mechanism is particularly relevant on the immigrant-dense side of the frontier, where reliance on co-ethnic networks rather than networks connected to the societal majority may reinforce economic isolation and limit exposure to higher-paying opportunities (Dustmann et al. 2010). These constraints affect minority residents, but they may also disadvantage majority-group residents living on the minority-dense side of the frontier. Although such residents may retain access to broader majority co-ethnic networks outside the neighbourhood, their local access to same-group ties and institutions can be thin. This can reduce engagement with schools and parent networks through which employment-relevant information circulates and, importantly, produce a form of dual network disadvantage (e.g., Small 2009).

SF further heighten territorial perceptions and asymmetrical control. Majority groups may perceive the frontier as a mechanism for maintaining social distance and preserving cultural dominance (Alba 2005), while minority groups may view it as an obstacle to integration and economic progress (Iyer and Pryce 2024). Sharp boundaries heighten awareness of group differences and may evoke feelings of threat or exclusion (Legewie and Schaeffer 2016). Such perceptions can reduce individuals' willingness to cross neighbourhood boundaries for work or services, effectively narrowing the spatial scope of job search. Reduced spatial mobility, in turn, diminishes employment opportunities and the quality of job matching (Zenou 2011; Manning and Petrongolo 2017; Severinsen et al. 2025).

Frontier proximity increases exposure to stark socio-economic contrasts, heightening relative deprivation among lower-status groups and producing status dissonance or a perceived loss of expected cultural or social centrality among majority outsiders (Luttmer 2005; Nieuwenhuis et al. 2017; Ridgeway 2019). These processes can weaken aspirations and reduce engagement with employment. They also intersect with disruptions to within-group status structures. Social identity theory suggests that individuals maintain self-esteem partly through favourable within-group comparisons and the maintenance of valued group status (Hogg 2016). Frontier proximity complicates these comparisons. Minority residents face daily exposure to higher-status out-groups, while majority outsiders find their expected status unrecognised in the local context. Research on stigma consciousness and stereotype threat indicates that such disruptions can reduce educational engagement and occupational ambition (Pinel 1999; Santiago et al. 2024). SF also heighten the salience of cultural identity. Minority groups may pursue cultural preservation strategies that do not necessarily align with labour market opportunities (Bisin and Verdier 2000). The majority of residents living on the minority-dense side may experience symbolic threat and withdraw from local institutions, further reducing opportunities for cross-group interaction (Stephan and Stephan 2000).

These dynamics are gendered. Research consistently shows that women's activity spaces tend to be more locally bounded than men's (Hanson and Pratt 1995; Schwanen et al. 2008; Osland 2010), and that gender norms shape access to public space, social networks and labour market pathways (Ridgeway and Correll 2004; Giulietti et al. 2013). Minority women residing on the majority side of the frontier often have more locally bounded activity spaces and may face stronger constraints arising from gendered norms. Limited access to same-group networks may therefore be more consequential for them than for minority men, who tend to exhibit greater spatial mobility and wider exposure to weak ties beyond the neighbourhood (McDowell 2008; Ryan 2011).

More generally, minority men—including those residing on the majority side of the SF may face higher levels of labour market discrimination and stereotyping, as well as stronger pressures to conform to majority norms concerning work and social behaviour. These pressures can generate identity stress and reduce their ability to benefit from bridging opportunities (Bertrand and Mullainathan 2004; Carlsson and Rooth 2007). Among the majority of residents living on the minority-dense side, women tend to report higher levels of perceived insecurity and vulnerability in unfamiliar or marginalised environments. This can reduce engagement with neighbourhood institutions such as schools or parent networks, which are key arenas for building both bridging and bonding ties (Pain 2000; Lareau 2011). Majority men embedded in neighbourhoods where they lack local dominance may respond more strongly to perceived losses of status or symbolic control in the local context, contributing to territorial withdrawal, reduced trust and avoidance of cross-group interaction. Their typically weaker institutional embeddedness within the neighbourhood further restricts access to locally mediated job information (Sampson 2012; Enos 2017; Laurence and Bentley 2016).

Overall, heterogeneous impacts are likely to be important. Within each SF neighbourhood, local homogeneity is disrupted by 'outsiders', such as native residents living in immigrant-dense areas or immigrants living in native-dense areas. These outsiders may possess fewer social networks, positive role models or relevant peer influences than the local majority, which may further constrain their access to information about employment opportunities. Consequently, both the magnitude and the direction of earnings effects are likely to vary across these groups.

Taken together, reduced direct and indirect contact, weakened bridging capacity, territorial threat perceptions, heightened

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relative deprivation, destabilised status structures and intensified cultural identity dynamics constitute central mechanisms embedded within SF neighbourhood environments. These mechanisms form a mutually reinforcing system that shapes labour market behaviour and opportunity structures. They affect both minority residents and majority outsiders, albeit in distinct ways and provide a comprehensive framework for understanding how proximity to SF may influence employment outcomes and long-term income trajectories.

3 | Basic Hypotheses

Previous theoretical and empirical research show that there could be distinct and complex dynamic processes and mechanisms prevailing on each side of the SF which may impact labour market outcomes. So, imagine we have identified one neighbourhood with a relatively high proportion of *immigrants*, and a spatially contiguous neighbourhood where the proportion of *natives* is substantially higher.

Our first research question is whether the earnings for those living on the high ethnic minority side of the SF (SF-High) would have been significantly better compared to an imaginary situation, where these people had lived on the native dense or low-immigrant-proportion side (SF-Low). This research question is related to whether the SF side matters, and it *does not* relate to the impact of the SF per se, as individuals living on both sides are exposed to the frontier.

The second research question is whether individuals living in an SF-High area would have had significantly different earnings compared to an imaginary situation where they had lived in a neighbourhood with similar immigrant density, but a neighbourhood *without* any SFs.

In treatment effects terminology, we consider exposure to the SF to be the treatment of interest, and we estimate the (average) effect of the SF-High treatment for the treated. For the second research question, we estimate the (average) impact of SF-High living as opposed to the counterfactual scenario of living in a similar neighbourhood, but with no adjacent SFs.

Moreover, based on results in the empirical literature on neighbourhood effects (G. Galster et al. 2010; Santiago et al. 2024), we stratify our testing on both gender and immigrant background to study the related heterogeneity. Important reasons for studying segregation along ethnic lines is that this is the main focus in the general literature, and immigrants are frequently viewed as one of the most vulnerable groups when it comes to attachment to labour markets, this is so in particular for immigrant women (Horgen 2014).

In the following, we present and discuss the empirical setting, the identification of the SFs, as well as our strategy for causal inference.

4 | Empirical Strategy

Our analysis is based on rich individual-level administrative records provided by Statistics Norway (SSB). These data originate from various national registers, for example, the National Population register. In Norway, all residents have a unique identification number, which is used in transactions between the individual and the public or private sector; examples include salary, tax payment and unemployment insurance. This information comes along with detailed information characterising the population (e.g., gender, birthdate, country background). The main advantages are that it is possible to follow the entire population over a long period of time, information have high reliability, and there are no problems of attrition.

4.1 | Neighbourhood Structure and SF-Identification

Our population consisted of everyone living in Oslo, the Norwegian capital, in January 2010 ($n = 584{,}631$). Overall, the populated area is 147 km^2. In our analysis, we focus on individuals in their prime working ages (18 and 59 years, $n = 326{,}718$). The neighbourhoods are defined as SSB′s, Basic Statistical Units (BSU, $n = 551$) with a strictly positive population. Their geographical demarcation has been developed by SSB for statistical purposes and is stable over time. They represent the smallest administrative unit, with an average size of 0.27 km^2. Moreover, they are spatially coherent and specifically intended to represent homogeneous areas across a range of social, demographic and functional dimensions (Byfuglien and Langen 1983), with an aim of creating zones that mimic natural neighbourhoods. Hence, these administrative boundaries which have existed since 1980 (op. cit.) are useful when delineating neighbourhoods. However, despite this, it is impossible to rule out that SFs may exist in other places than along the administrative boundaries that we use.

Because the neighbourhoods are located within the same municipality, the inhabitants are likely to benefit from a similar quality of public services (e.g., free public schools, municipal language training for immigrants or employment services). Still, there may be differences in the availability, size and quality of housing and surrounding amenities that, together with other subtle differences in the quality of welfare services as well as individual preferences, cause residential sorting, which over time leads to differences in the composition of inhabitants.

In Figure 1, we present stylised examples of SF neighbourhoods. On either side of directly adjacent neighbourhoods, demographic or social characteristics of the populations may differ. If there is a clear spatial discontinuity, an SF between the neighbourhoods could exist. In this paper, we study spatial discontinuities in terms of demographic characteristics—namely, immigrants versus natives.[1] Many neighbourhoods have no SF, but those that do are identified by having different population origin densities.

A 'low' SF neighbourhood (SF-Low) is defined as a neighbourhood with a statistically significant and substantially lower immigrant and descendant population proportion compared to the adjacent neighbourhood. This definition then implies that the adjacent neighbourhood will be defined as an SF-High. This is illustrated in Example 1 in Figure 1. In Example 2 of Figure 1, a different scenario arises. Neighbourhood B is both an SF-High neighbourhood along its western border and an SF-Low neighbourhood along its southern border. In such a case, the neighbourhood is labelled SF-Mixed. Mixed SFs are not included as an SF in this analysis.

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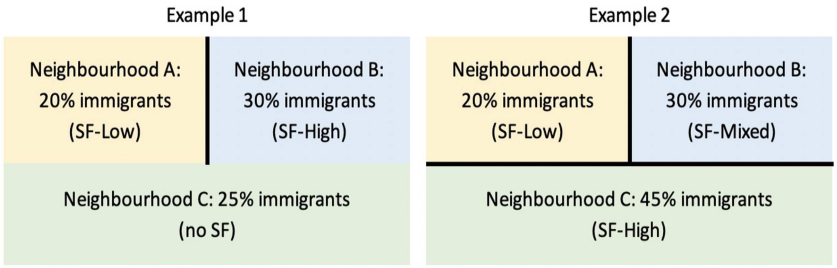


FIGURE 1 | Stylised social frontiers.*Source:* Authors' own work.

TABLE 1 | Neighbourhood population, by social frontiers.

				Ages 18–59			
Social frontiers	**Total # nbh.**	**# obs.**	**# nbh.**	**# obs.**	**Native**	**Immigr.**	**Mixed**
None	338	357,120	337	228,952	148,168	56,296	24,488
High	86	99,534	86	61,274	27,257	27,436	6581
Low	114	112,689	114	68,786	48,030	13,756	7000
Mixed	13	15,288	13	9201	5017	3156	1028
Total	551	584,631	550	368,213	228,472	100,644	39,097

Note: Oslo population, January 2010. Citizens of 'mixed' origin have two immigrant parents (descendants) or parents of mixed origin, for example, one native and one immigrant. The 'descendants' make up about 1/4 of the total group of persons with 'mixed' background. Source: SSB and own calculations.

To identify SFs, we follow Dean et al. (2019)'s two-step procedure for locating spatial discontinuities. This implies using a Bayesian local spatial conditional adaptive auto-regressive model for a binomial response variable based on Lee and Mitchell (2013). In standard spatial models, the $n \times n$ spatial weight matrix is commonly fixed with values of 0 and 1, where 1 denotes contiguous zones and 0 refers to non-contiguous zones. To identify the SF, which are contiguous yet dissimilar, we use an iterative algorithm applying a modified asymmetric spatial weights matrix so that spatially adjacent zones can have a weight of 0 (Dean et al. 2019). Hence, we allow the two contiguous areas to have a step change in the proportions of immigrants. Moreover, we limit the sample to boundaries having a statistically significant (5% level) difference.

Secondly, to avoid selecting statistically significant proportions which do not represent substantive step changes, we limit our SF-sample to borders which also have substantially larger immigrant/native proportions (outside the 95% confidence interval). So, in the case of statistically significant and substantial step changes in contiguous immigrant proportions, an SF boundary has been identified. Of the 551 neighbourhoods, 114 were SF-Low's (14% immigrants on average, see Table 2), and 86 were SF-High's (36% immigrants). This distribution is shown in Table 1. In Figure 2, a map illustrates the SF along the borders of BSUs in Oslo.

4.2 | Description of Sample and Matching Procedures

For our analysis, we restrict the total 2010 Oslo-population of 584,631 individuals to the following sample:

- Pre-tax annual earned income below NOK 2 m ($n = 581{,}587$),
- Age 18–59 years ($n = 365{,}563$),
- Immigrants or natives (not 'descendants' or 'mixed' origins, $n = 326{,}718$),
- Living in either of the following: an SF-High ($n = 54{,}613$), an SF-Low ($n = 61{,}470$) or a non-SF with a share of immigrants *comparable to the SF-High* neighbourhoods—we discuss the definition of this below.

We focus on inhabitants in SF-Low and SF-High, as well as a third type of neighbourhood that we label 'BSU-High'. The latter category of neighbourhoods is a non-SF neighbourhood with an immigrant share comparable to the SF-High neighbourhoods. In total, 51 BSU-Highs are chosen by selecting the nearest neighbour in terms of the neighbourhood proportion of immigrants. Figure 3 shows the immigrant share across all 551 Oslo neighbourhoods ordered from low to high, as well as where in the distribution the SF-High, SF-Low and SF-Mixed neighbourhoods are located. The BSU-High neighbourhoods are primarily located in the right tail of the distribution.

Table 2 shows the population characteristics of the neighbourhoods according to the SF-High and SF-Low classification. For comparison, we have included all 551 Oslo neighbourhoods. Each neighbourhood is given equal weight, irrespective of population size. For SF-High, the statistics show that there are fewer women, and the average age is almost 2 years younger compared with SF-Low. The highest completed level of education and earned incomes are substantially lower in SF-High; the share with very low earned incomes is almost 8 percentage points higher in SF-High, and the share with very high incomes is almost half in SF-High compared to SF-Low. The SF-High proportion of single-person households is 33% versus 26% in SF-Low, and the proportion of couples with children is 37% in SF-High versus 44% in SF-Low. The building mass reflects these differences: SF-Highs typically have large building blocks with multiple storeys and rented apartments, while the SF-Lows have typical Norwegian privately owned detached housing. In

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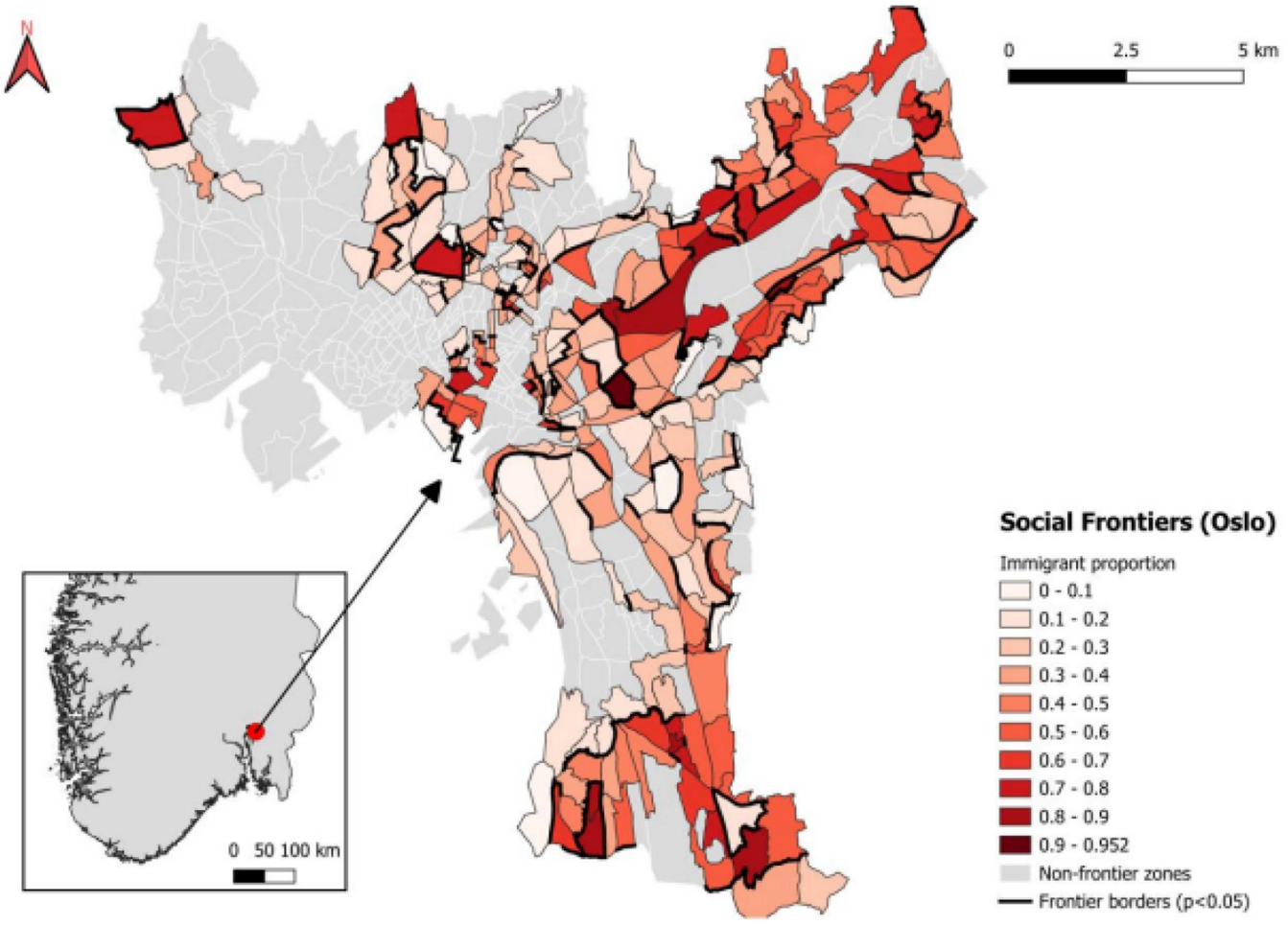


FIGURE 2 | Map of social frontiers in the study area. *Note:* The map shows the identified SF along the borders of BSUs in the municipality of Oslo. BSUs with SF are classified according to immigrant proportions. The map on the left shows the study area within Norway.*Source:* Authors' own work.

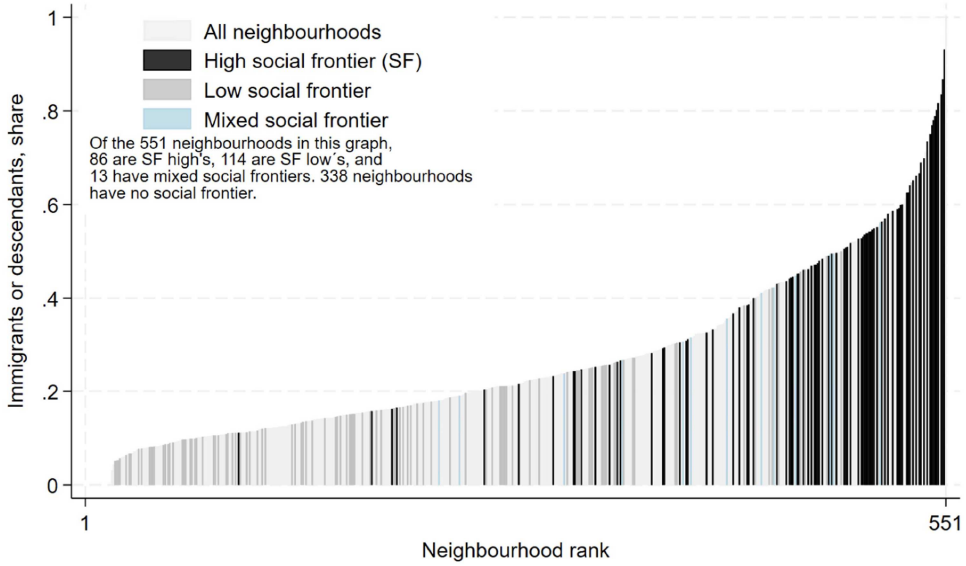


FIGURE 3 | Neighbourhood immigrant distribution.*Source:* SSB and own calculations.

addition, the homes in SF-High are somewhat smaller, although there are an equal number of persons in the household. The building mass is on average almost 6 years younger in the SF-Highs compared to the SF-Low.

4.3 | Individual-Level Background Information

We use information about the individual such as gender, age, country background, entry to Norway for immigrants and the

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TABLE 2 | Population summary, by SF-type.

	SF-High	SF-Low	Total
Origin: Native	45.3	**73.3**	**64.9**
Immigrant	36.4	**14.1**	**21.0**
Descendant	12.1	**3.8**	**5.5**
Mixed	6.2	**8.7**	**8.6**
Non-Western origin	36.0	**11.8**	**17.3**
Women	45.8	**50.7**	**49.0**
Age[#]	37.0	**39.0**	**38.2**
Younger than 18 years	19.6	18.6	**17.4**
Older than 59 years	15.6	**18.3**	**17.5**
Enroled in education*	28.1	27.1	26.6
Education: none or unknown	25.1	**20.0**	**20.6**
Lower secondary	26.2	**19.2**	**19.3**
Upper secondary	26.1	26.7	25.2
University or college, lower level	16.2	**22.1**	**22.4**
University or college, higher level	6.4	**12.0**	**12.5**
Low or no education, age 18−59	38.8	**23.1**	**25.9**
Higher education, age 18−59	30.7	**47.3**	**45.9**
Earned income, age 18−59[#]	304.4	**413.6**	**403.1**
Low earned income, age 18−59	24.8	**17.3**	**19.0**
High earned income, age 18−59	4.7	**10.4**	**10.6**
Household: single person	33.2	**25.6**	32.0
Couple, no children	13.4	**18.2**	**17.1**
Couple, with one or more children	36.8	**44.2**	38.1
Single parent	8.3	**5.7**	**6.3**
Other family type	8.3	**6.3**	**6.6**
Building type: 3−4 storeys	30.1	**23.4**	26.0
Building with 5 or more storeys	25.9	**15.3**	23.3
Detached single-family house	11.0	**17.7**	**14.8**
Detached two-family house	5.0	**11.7**	**8.2**
Semi-detached house	4.7	**12.6**	**7.2**
Other type of building or unknown	23.3	19.3	20.4
Home ownership: owner	32.6	**41.7**	**43.9**
Co-operative ownership	31.7	35.3	27.9
Rental	31.2	**20.3**	**24.5**
Other or unknown ownership	4.5	**2.7**	3.8
Number of rooms[#]	3.3	**3.7**	**3.5**
Utility floor space, $m^{2\#}$	92.1	**104.0**	**101.9**
Persons in family[#]	2.6	2.6	2.5
Construction year[#]	1966.8	**1961.0**	**1958.6**
Neighbourhood inhabitants[#]	1157.4	988.5	1061.0
Neighbourhoods[#]	86	114	551
Inhabitants[#]	99,534	112,689	584,631
Inhabitants, age 18−59[#]	61,274	68,786	368,213

Note: (*) In education is equal to one for everyone 15 years or younger. (#) indicates a continuous variable. All variables are unweighted means of neighbourhood averages. A bold font indicates a statistically significant difference at the 0.05-level relative to SF-High. Low earned income below 1 G; high above 10 G. 1 G equals NOK 75,641 in 2010. Annual earned income before taxes. Source: SSB and own calculations.

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neighbourhood of residence. Using an anonymous personal identifier, we merge information about individuals and the type of housing where the family resides. For the years from 2008 to 2018 (both years included), we have information on individuals' pre-tax earned incomes. Tables 3 and 4 shows selected characteristics of Norwegian and immigrant men and women. The raw data show strong evidence of residential sorting, especially comparing SF-High and SF-Low.

We construct nine outcome variables: one for each year from 2010 to 2018. The 2010 outcome is equal to the individual earnings for that year, while the 2011 outcome is the cumulative earnings for 2010 and 2011, and so on. All amounts are measured in thousands of CPI-adjusted Norwegian kroner (NOK). Annual earnings for 2008 and 2009 are used as pre-treatment control variables (see Tables 5A and 5B).[2]

4.4 | Causal Identification Strategy

In the absence of experimental evidence, we rely on rich observational data and apply PSM to address residential sorting (Caliendo and Kopeinig 2008). The idea behind matching is to mimic the properties of randomisation. If treated and control individuals are alike in every relevant pre-treatment characteristic, any subsequent difference in earnings can be attributed to the treatment. While randomisation ideally ensures causal identification unconditional of individual characteristics, matching relies on the assumption of *selection-on-observables*. In other words, we assume that residential sorting is based on observed characteristics.

PSM is implemented using an algorithm that matches each treated individual to the nearest control unit. Here, 'nearest' refers to the distance between two propensity scores (see Rosenbaum and Rubin 1983), which, in this case, represent the estimated probability of living in an SF-High neighbourhood.

PSM offers two main advantages over other non-experimental evaluation methods. First, it is non-parametric, meaning it does not impose a functional form on the relationship between covariates and outcomes. Second, it highlights the common support problem—a lack of overlap in the propensity scores between treated and control units, which can be a source of bias (Heckman et al. 1999). However, PSM requires both a sufficient number of control observations and relevant pre-treatment information. A larger pool of potential controls increases the

TABLE 3 | Sample summary statistics for immigrants, by SF-type.

	Men			Women		
	SF-High	**SF-Low**	**BSU-High**	**SF-High**	**SF-Low**	**BSU-High**
Non-Western origin	80.9	**59.3**	**73.1**	84.9	**68.2**	**80.2**
Age[#]	38.1	**38.7**	**37.8**	36.4	**37.4**	**36.0**
Enroled in education	10.1	**6.2**	**7.3**	13.3	**10.6**	**10.3**
Low or no education	50.2	**42.3**	51.5	54.5	**42.6**	53.8
Higher education	22.3	**29.8**	22.0	23.7	**35.9**	23.8
Earned income[#]	286.7	**339.5**	288.7	191.5	**250.6**	**198.8**
Low earned income	24.4	**20.4**	24.7	38.8	**29.4**	38.2
High earned income	2.9	**6.0**	3.4	0.9	**2.3**	1.2
Household: single person	37.2	37.2	37.9	18.3	**20.2**	**19.6**
Couple, no children	9.4	**13.1**	**10.3**	12.2	**16.1**	**13.2**
Couple with one child or more	41.4	**38.3**	**36.5**	46.6	**43.8**	**42.7**
Single parent	1.9	**1.4**	2.0	11.9	**9.6**	**12.9**
Other household type	10.1	10.1	**13.3**	11.1	10.2	11.5
Building type: 3+ storeys	65.5	**49.1**	62.7	68.8	**52.0**	67.7
Detached housing	11.5	26.7	**11.9**	12.7	**29.6**	**13.2**
Other or unknown building type	23.0	24.2	**25.3**	18.4	18.5	19.1
Home ownership: owner	21.7	**26.7**	**25.6**	23.8	**29.6**	**27.0**
Co-operative	36.6	**33.7**	**27.9**	40.3	**38.5**	**31.0**
Rental	35.0	**32.7**	**38.9**	29.6	**26.2**	**35.7**
Other or unknown ownership	6.6	6.9	**7.6**	6.3	5.7	6.2
Number of rooms[#]	3.0	**3.2**	**2.9**	3.1	**3.3**	**3.1**
Utility floor space, $m^{2\#}$	81.5	**90.1**	81.0	85.1	**94.8**	**82.9**
Persons in family[#]	2.7	**2.5**	**2.5**	3.1	**2.9**	**3.0**
Construction year[#]	1962.2	**1961.8**	**1966.6**	1971.1	**1965.3**	**1969.9**
Individuals, *N*	14,298	7081	7121	13,127	6655	6301

Note: Subject to sample criteria. (#) denote a continuous variable, other variables in percentages. Bold fonts indicate significance at the 0.05-level compared to SF-High. BSU-High denotes 51 non-frontier neighbourhoods with immigrant proportions similar to SF-High. Low earned income below 1 G; high above 10 G. 1 G equals NOK 75,641 in 2010. Annual earned income before taxes. Source: SSB and own calculations.

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TABLE 4 | Sample summary statistics for natives, by SF-type.

	Men			Women		
	SF-High	**SF-Low**	**BSU-High**	**SF-High**	**SF-Low**	**BSU-High**
Age[#]	38.9	38.9	**37.3**	39.2	**38.9**	**36.7**
Enroled in education	12.5	11.8	12.3	15.1	14.9	**17.8**
Low or no education	27.1	**17.8**	**21.8**	22.0	**13.8**	**17.7**
Higher education	36.9	**50.3**	**47.9**	45.6	**59.2**	**54.5**
Earned income[#]	419.9	**512.9**	**442.7**	347.5	**399.7**	352.9
Low earned income	16.3	**11.8**	**13.9**	15.9	**12.1**	15.2
High earned income	11.3	**18.9**	**12.9**	3.7	**6.3**	4.2
Household: single person	42.9	**31.2**	**49.9**	33.2	**27.0**	**42.8**
Couple, no children	15.4	**16.5**	**18.7**	17.0	17.0	**19.5**
Couple with one child or more	34.4	**45.5**	**22.7**	35.8	**44.9**	**25.7**
Single parent	2.4	2.4	**1.3**	9.1	**7.0**	**6.6**
Other household type	4.8	4.4	**7.8**	4.9	**4.2**	5.5
Building type: 3+ storeys	61.4	**45.8**	**63.3**	62.9	**46.2**	**64.3**
Detached housing	24.4	**40.8**	**20.8**	24.6	**40.7**	**22.9**
Other or unknown building type	14.2	**13.4**	**15.9**	12.4	13.0	12.9
Home ownership: owner	36.9	**44.3**	**40.4**	38.0	**43.6**	**40.9**
Co-operative	39.0	39.2	**29.8**	42.1	**40.7**	**32.3**
Rental	22.8	**15.6**	**28.5**	19.1	**14.8**	**25.7**
Other or unknown ownership	1.2	**0.9**	1.3	0.8	0.9	**1.1**
Number of rooms[#]	3.3	**3.6**	**3.1**	3.3	**3.6**	**3.1**
Utility floor space, $m^{2\#}$	92.0	**104.7**	**85.1**	93.2	**104.9**	**87.2**
Persons in family[#]	2.2	**2.5**	**1.8**	2.3	**2.5**	**2.0**
Construction year[#]	1966.9	**1960.8**	**1957.2**	1967.4	**1960.4**	**1956.5**
Individuals, *N*	13,820	23,243	8471	13,368	24,491	8029

Note: Subject to sample criteria. (#) denote a continuous variable, other variables in percentages. Bold fonts indicate significance at the 0.05-level compared to SF-High. BSU-High denotes 51 non-frontier neighbourhoods with immigrant proportions similar to SF-High. Low earned income below 1 G; high above 10 G. 1 G equal to NOK 75,641 in 2010. Annual earned income before taxes. Source: SSB and own calculations.

TABLE 5A | Raw cumulative earnings (2010–18) for immigrants. NOK 1000.

	Men			Women		
Year	**SF-High**	**SF-Low**	**BSU-High**	**SF-High**	**SF-Low**	**BSU-High**
2008*	261.5	303.2	253.1	163.1	216.0	159.2
2009*	275.4	329.3	275.5	179.7	239.3	183.6
2010	283.6	337.0	285.9	190.0	249.8	197.0
2011	579.6	685.7	580.0	390.2	511.9	403.9
2012	889.0	1045.5	882.3	602.4	786.8	620.6
2013	1200.6	1409.7	1187.7	820.2	1069.6	842.6
2014	1509.8	1772.4	1489.5	1041.1	1356.7	1065.3
2015	1839.5	2148.5	1812.6	1282.7	1660.6	1306.4
2016	2162.7	2520.4	2127.4	1523.9	1963.7	1545.9
2017	2489.0	2895.6	2440.9	1769.6	2269.0	1786.7
2018	2814.1	3267.4	2752.0	2019.0	2573.0	2029.5

Note: (*) 2008 and 2009 are average annual pre-tax earned incomes. 2010–18 are cumulative pre-tax earned incomes; NOK 1000, CPI-adjusted, base 2015. BSU-High denotes 51 non-frontier neighbourhoods with immigrant proportions similar to SF-High. Source: SSB and own calculations.

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TABLE 5B | Raw cumulative (2010–18) earnings for natives. NOK 1000.

	Men			Women		
Year	**SF-High**	**SF-Low**	**BSU-High**	**SF-High**	**SF-Low**	**BSU-High**
2008*	395.4	478.1	408.5	321.0	368.6	319.8
2009*	405.1	490.8	424.9	334.7	383.5	336.8
2010	414.4	505.2	439.4	342.9	393.8	349.7
2011	849.1	1036.4	903.4	701.6	806.3	716.5
2012	1302.5	1592.8	1389.1	1076.8	1237.2	1100.7
2013	1766.3	2162.1	1890.0	1460.5	1679.4	1493.6
2014	2234.6	2744.0	2400.6	1849.1	2130.3	1895.3
2015	2733.2	3347.1	2935.5	2273.1	2608.3	2326.4
2016	3226.1	3944.0	3469.4	2694.5	3085.2	2765.9
2017	3724.5	4546.2	4009.7	3119.2	3567.5	3191.8
2018	4222.7	5151.0	4559.7	3542.4	4050.8	3630.9

Note: (*) 2008 and 2009 are average annual pre-tax earned incomes. 2010–18 are cumulative pre-tax earned incomes; NOK 1000, CPI-adjusted, base 2015. BSU-High denotes 51 non-frontier neighbourhoods with immigrant proportions similar to SF-High. Source: SSB and own calculations.

likelihood of finding good matches, thereby improving covariate balance.

As discussed, to account for residential sorting, we use demographic and socioeconomic administrative data to calculate the probability that a given individual would live in an SF-High neighbourhood (i.e., the propensity score, or *p*-score). This probability enables us to identify a statistical twin–an individual who did not sort into an SF-High neighbourhood–who can serve as a control. Since pre-treatment earnings are likely correlated with unobserved variables (e.g., ability), controlling for prior earnings can implicitly account for unobserved determinants of neighbourhood sorting. For further details on the PSM procedure, we refer the interested readers to the Supporting Information Materials.

5 | Results

We now present the results. First, we discuss the comparison between SF-High and SF-Low, followed by the SF-High versus BSU-High scenario. In the latter, the counterfactual scenario involves living in a high-immigrant-density neighbourhood without an SF. In both cases, the outcome of interest is earned income from 2010 to 2018, with 2010 serving as the year of 'pseudo-randomisation'. For each post-treatment year, we analyse the cumulative average of gross annual earnings from 2010 up to that year. For example, the outcome for 2015 represents the sum of annual earnings from 2010 to 2015.

5.1 | SF-High Versus SF-Low

For the SF-High versus SF-Low scenario, the cumulative earnings outcomes from 2010 to 2018 are presented as a line in Figure 4, while the average treatment effect on the treated (ATT)–the difference between the two groups–is plotted in Figure 5. The grey left part of Figure 4 includes scatterplots showing the average earned incomes for each pre-treatment year (2008 and 2009) separately. These pre-treatment outcomes illustrate the balancing properties of the matching procedure: there is only a small bias in average pre-treatment earnings between the on-support treatment group (marked 'x' in the figure) and the matched controls (marked 'o' the figure). In Table 6, we present bias-corrected ATT estimates, accounting for these pre-treatment differences.

For immigrant men (upper left panel), the average earnings in 2010 were approximately NOK 293,000 for the treatment group and NOK 301,000 for the control group. The cumulative average earnings from 2010 to 2018 were just below NOK 2.89 million for the treated and NOK 3.02 million for the controls. That is, the counterfactual scenario—the dashed line—shows slightly higher average earnings. This suggests a negative treatment effect for immigrant men living in an SF-High neighbourhood, compared to what they would have earned if they lived on the other side of the 'cliff edge'.

Figure 5 displays the ATT alongside a 95% confidence interval. For immigrant men, the cumulative difference in earnings from 2010 to 2018 amounts to *minus* NOK 129,000–an annual earnings *penalty* of NOK 14,400. The pre-treatment difference (in Figure 4) indicated an initial bias of *minus* NOK 10,200 in disfavour of the treatment group. Assuming this bias remains constant over time, adjusting the ATT for the bias reduces the SF-High *penalty* to NOK 4200 per year on average.

For immigrant women (upper right panel), the story is similar, but we note that their cumulative earnings are considerably lower, topping at an average of NOK 207 million in 2018; the ATT in 2018 is *minus* NOK 76,000, which is statistically insignificant from zero. This is a *penalty* of NOK 8500 per year or NOK 6200 bias-corrected. For native men (lower right panel), the ATT is *minus* NOK 145,200 in 2018, which is statistically significant during the entire period. The estimate is equivalent to an average *penalty* of NOK 16,100. The pre-treatment bias is estimated at *minus* NOK 7300, which indicates a bias-corrected treatment effect of *minus* NOK 8800 per year. Finally, for native women (lower left panel) the treatment effect is statistically insignificant through all years and the point estimate in 2018 is *minus* NOK 27,200. With an indicated pre-treatment bias of around minus NOK 1100, the biased-corrected treatment effect is *minus* NOK 1900 per year.

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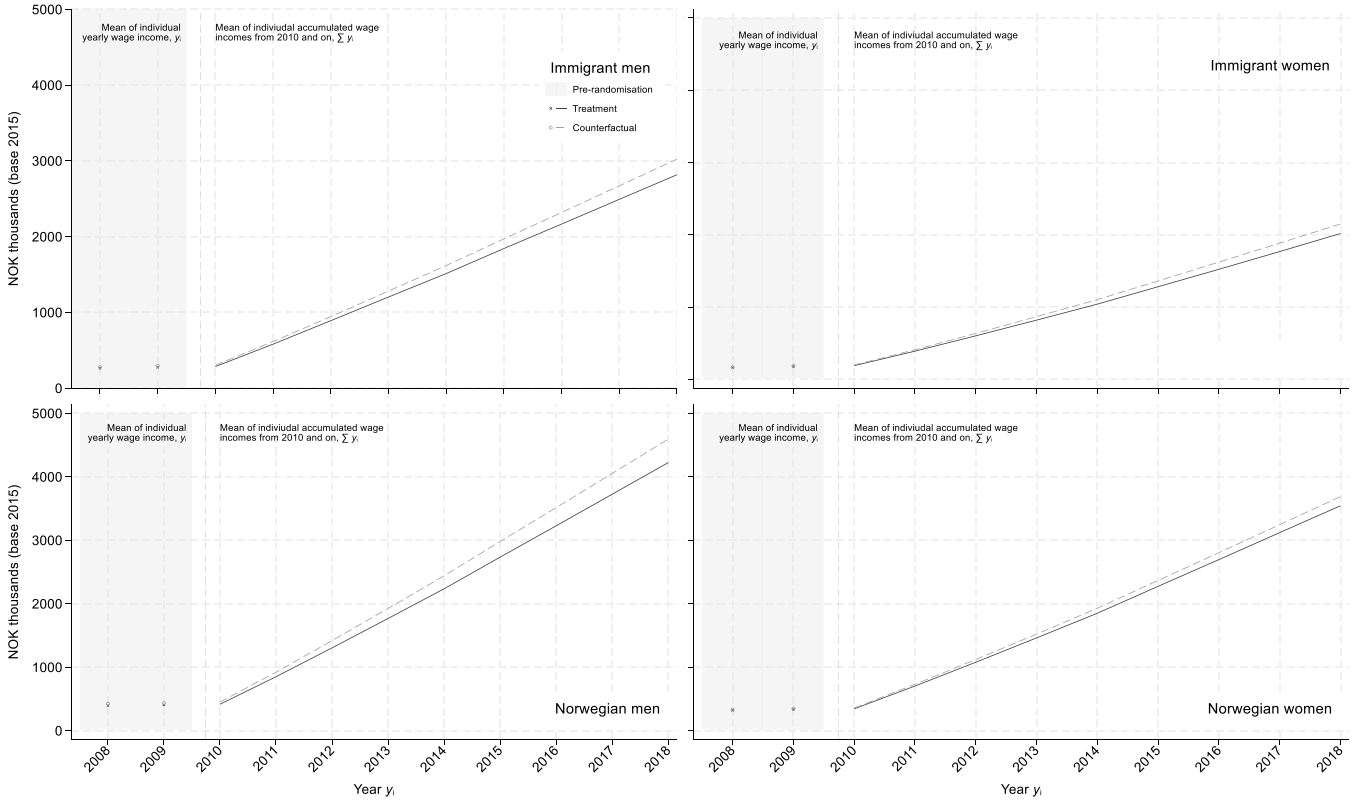


FIGURE 4 | SF-High versus SF-Low, outcome in levels (2010–18). Note: Mean cumulative earnings for treated and matched controls (lines) and mean of pre-treatment outcomes (scatter). *Source:* Own calculations based on administrative data from SSB.

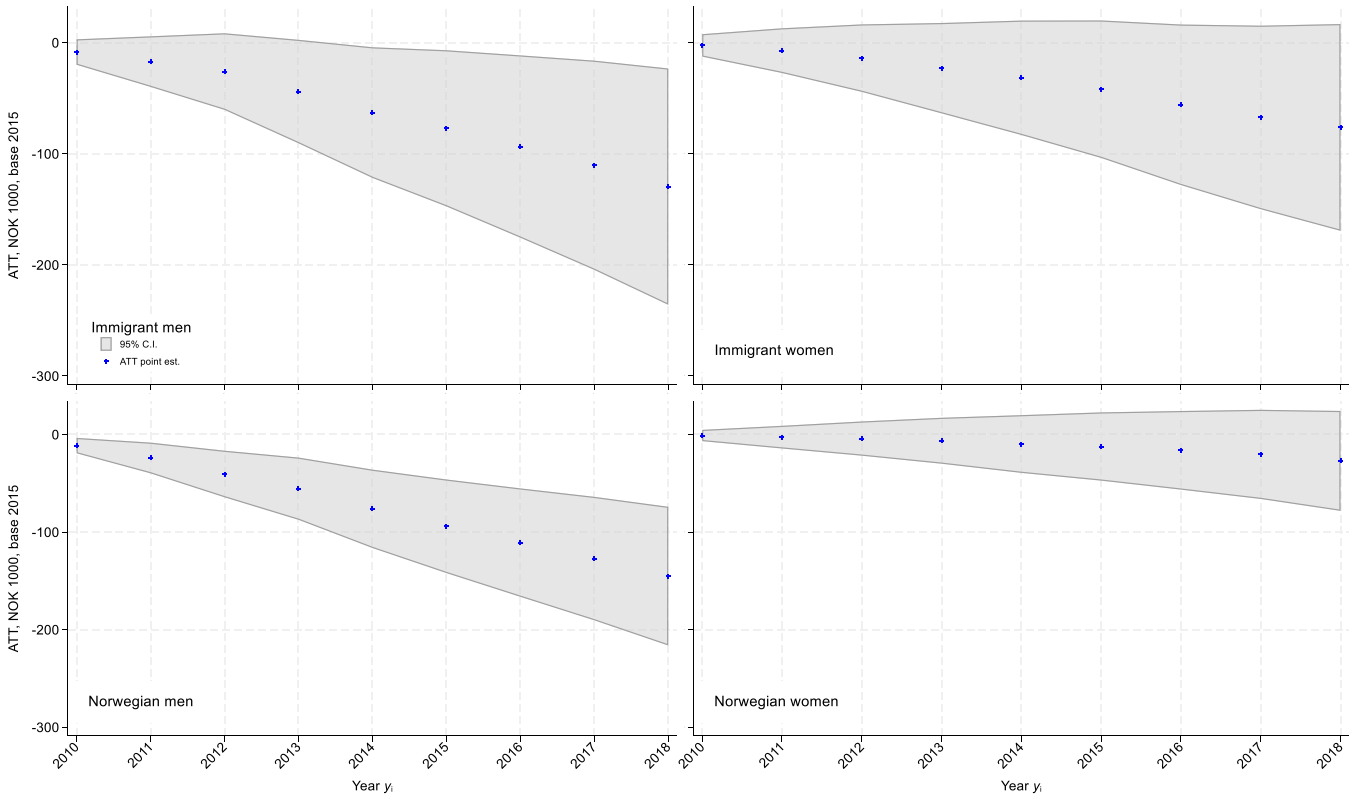


FIGURE 5 | SF-High versus SF-Low, average treatment effect on the treated (ATT). Note: The ATT is the difference in mean cumulative earnings for treated and matched controls; CPI-adjusted, base 2015, NOK 1000. The CI-band increases over time due to increasing cumulative earnings. *Source:* Own calculations based on administrative data from SSB.

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TABLE 6 | Main results, SF-High versus SF-Low. Amounts in NOK 1000.

Subgroup	Treated	Control	ATT	ATT/year	Bias	ATT adjusted
Immigrant men	2891.9	3021.1	**−129.2**	−14.4	−10.2	−4.2
Immigrant women	2073.7	2150.1	−76.4	−8.5	−2.3	−6.2
Native men	4447.2	4592.4	**−145.2**	−16.1	−7.3	−8.8
Native women	3659.2	3686.4	−27.2	−3.0	−1.1	−1.9

Note: Subject to rounding errors. The first column shows the cumulative earnings (2010–18) for treated on support, while the second column is for the matched controls. The third column is the average treatment effect on the treated (ATT). A bold font indicates statistical significance at the 0.05-level. The fourth column is the pre-treatment earnings difference in 2009 between treated and controls. The last column is the annual bias-corrected ATT. Source: Own calculations based on administrative data from SSB.

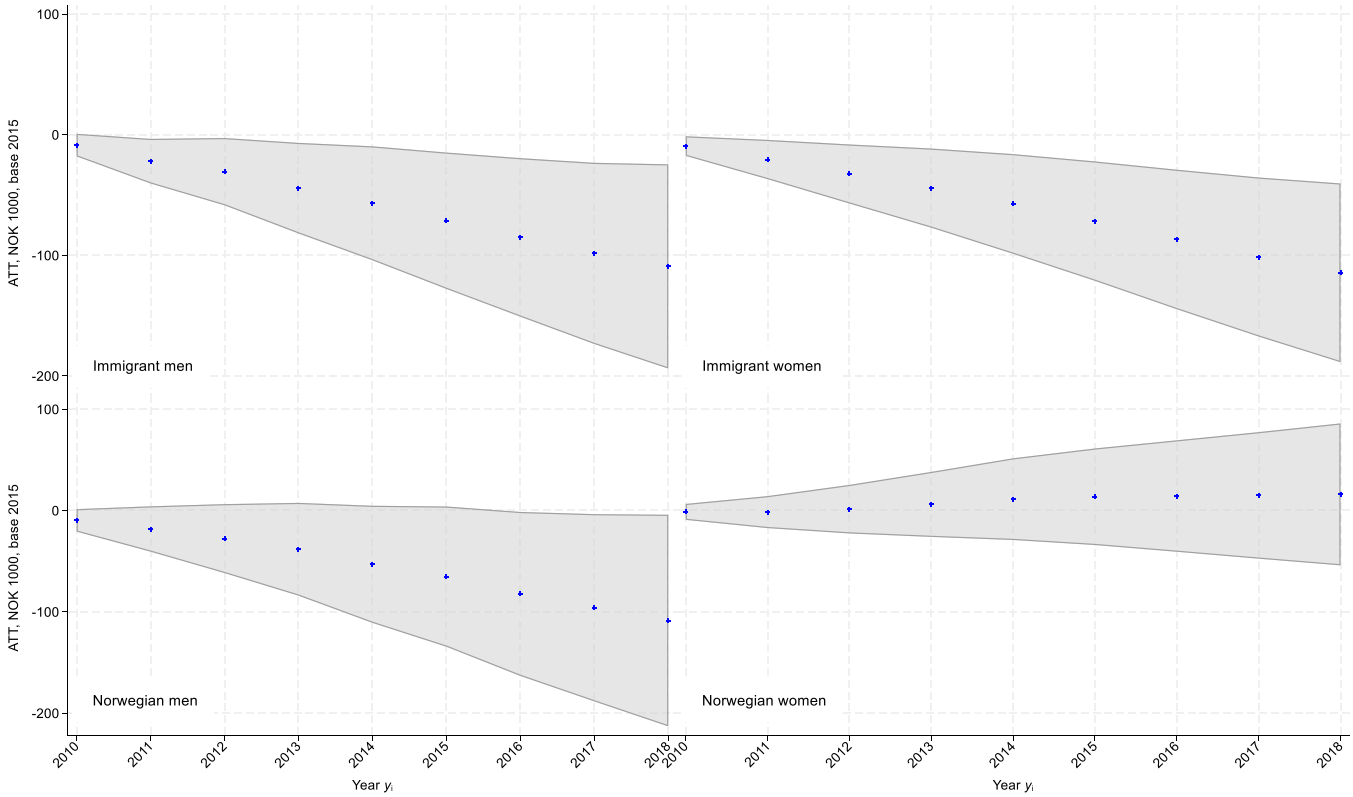


FIGURE 6 | SF-High versus BSU-High, average treatment effects (ATT). Note: The ATT is the difference in mean cumulative earnings for treated and matched controls; CPI-adjusted, base 2015, NOK 1000. The CI-band increases over time due to increasing cumulative earnings. *Source:* Own calculations based on administrative data from SSB.

Overall, the results indicate *negative* annual earnings effects of NOK 3000–8500 for women and NOK 14,400–16,100 for men. These effects were statistically significant for men, but not for women. For men, the estimated pre-treatment differences suggest bias-corrected point estimates ranging from NOK 4200 to 8800 per year, equivalent to approximately 1.3%–1.8% of the average annual earnings of a treated man. For women, the bias-corrected estimates range from minus NOK 6200 to minus NOK 1900, corresponding to an annual earnings loss of 0.5%–2.7%.

5.2 | SF-High Versus BSU-High

Figure 6 presents the ATT results for the SF-High versus BSU-High scenario. This represents a counterfactual scenario in which the neighbourhood has no SF but maintains an equally high immigrant density. In this analysis, the pool of possible controls includes all residents in neighbourhoods without frontiers. The matching algorithm selects the closest controls to balance both individual and neighbourhood characteristics.

For male immigrants (upper left panel), the result shows a significantly negative ATT of NOK 109,200 in 2018, or an annual *penalty* of NOK 12,100. As the bias in the pre-treatment outcome is estimated at minus NOK 10,900, the bias-corrected estimate of the per-year ATT is minus NOK 1200. Table 7 shows an overview of the main results; the levels are presented in Supporting Information S1: Appendix Figure A5. Here, the matching procedure largely includes the same variables as before, with only slight changes. First, as the frontier proportion is now undefined for the controls, this variable must be omitted. Second, as it is essential to balance the proportion of immigrants in the untreated neighbourhoods with that of the SF-High neighbourhoods, we have included the immigrant proportion. In the male immigrant treatment group, the immigrant proportion is 51.9%,

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TABLE 7 | Main results, SF-High versus BSU-High. Amounts in NOK 1000.

Subgroup	Treated	Control	ATT	ATT/year	Bias	ATT adjusted
Immigrant men	2834.9	2944.0	**−109.2**	−12.1	−10.9	−1.2
Immigrant women	2026.8	2141.4	**−114.6**	−12.7	−8.9	−3.8
Native men	4245.5	4345.3	**−108.7**	−12.1	−8.4	−3.7
Native women	3550.3	3534.5	15.8	1.8	> 0.1	1.8

Note: Subject to rounding errors. The first column shows the sum of earnings (2010−18) for treated, while the second column is for the controls. The third column is the average treatment effect on the treated (ATT). A bold font implies statistical significance at the 0.05-level. The fifth column is the estimated bias. The last column is the bias-corrected ATT per year. Source: Own calculations based on administrative data from SSB.

while it is 50.9% among the group of matched controls. The one percentage point difference is equivalent to a standardised bias percentage of 6.4. Among all covariates, the median (mean) bias is 4.3 (4.6)%. Overlap among treated and controls, as well as covariate balancing, is shown in Supporting Information S1: Appendix Figures A3 and A4, respectively.

For immigrant women, the estimated ATT is minus NOK 114,600, which is statistically significant; the bias is estimated at minus NOK 8900, and the bias-corrected ATT is minus NOK 3800 per year. For native men, the estimated numbers are very similar, although the earnings in levels are substantially higher compared to immigrants, thus reducing the importance in percentages of the earnings level. For native women, the estimated ATT is positive, but close to zero and statistically insignificant.

Overall, the SF-High versus BSU-High scenario shows slightly higher biases across covariates, although they remain within acceptable limits. Specifically, the neighbourhood proportion of immigrants appears to be balanced with differences ranging between 0.7 and 1.1 percentage-points in favour of the SF-High neighbourhoods. In summary, our choice of matching algorithm achieves a good covariate balance and a low bias in our pre-treatment outcomes.

In both scenarios, we find negative effects on earnings due to SF-High exposure. For immigrant and Norwegian men, as well as immigrant women in the BSU-High scenario, the ATTs are statistically significant. However, these effects are modest when compared to overall earnings levels. In summary, our evidence suggests that SF in Oslo, at worst, have had small negative consequences for the earnings of residents in frontier segregated, immigrant dense neighbourhoods. Additional regressions of the individual treatment effects on socioeconomic characteristics indicate that negative effects are most often associated with lower education levels, weaker labour market integration or single-person households.

In terms of treatment exposure, the average length of the SF accounted for 35% of the neighbourhood circumference for immigrants and 28% for natives. Among immigrants, 75% of men and 79% of women resided in the same neighbourhood in January 2009, compared to 27%–32% in 2004. For natives, 85%–86% lived in the same neighbourhood the previous year, while 46%–47% had done so 6 years earlier. We cannot rule out the possibility that higher treatment intensities would have led to more pronounced treatment effects.

6 | Discussion and Limitations

SF research connects closely to current themes in population geography by shifting attention from broad patterns of segregation to micro-geographies of boundaries. This boundary-focused perspective aligns with emerging work on scale and neighbourhood delineation, showing how inequality can be concentrated along edges even when larger zones appear mixed. It also complements ongoing topics on selective migration and neighbourhood change by treating frontiers as friction points that may shape who moves, who stays and how place reputations form.

Building on these links to population geography, this paper brings the SF lens into an earnings framework by asking whether living in an SF neighbourhood has measurable effects on income. We employ statistical methods to identify SF and use a causal design (PSM) to address self-selection into neighbourhoods. Regarding causal identification and the *selection-on-observables* assumption, our results provide empirical justification: pre-treatment earnings for the treatment group align closely with those of the matched controls, and our covariates are overall well balanced.

We test whether SF per se have an impact on earned income. The effects are highly similar for immigrant and native men, as well as for immigrant women, with the effect of an SF being negative and statistically significant in each case. For native women, however, the effect is small, positive and statistically insignificant. Regarding the effects of the frontier side, we find no significant penalty of living in an SF-High versus an SF-Low, either for immigrant or native women. For men of both native and immigrant origin, however, the effect is significant and negative. Overall, our findings document a statistically significant earnings penalty of the 2010 Oslo SF segregation, especially for men. However, the economic magnitude of these consequences appears modest.

Thus, like Olner et al. (2023), we find that the frontier side is important. To some extent, our results align with the findings presented by Severinsen et al. (2025), who report that native men living in SF-High are significantly less willing to commute, and to a larger extent, find their jobs within their residential neighbourhood. This may reduce the work potential and possible career paths for these native men. Our results could also be linked to the previously discussed characteristics of SF (Iyer and Pryce 2024), where inter-group contacts could be limited. Independent intra-group immigrant networks and amenities might be prevailing in SF-High which may neither be available nor relevant for native men. Along these lines, individuals could get a sense of low status and low self-esteem, accompanied by perceived deprivation in comparison to the majority group living in SF-Low. Such effects may have long-term consequences by reducing educational engagement and occupational ambition (Santiago et al. 2024). As mentioned in the literature review, exposure to inequality and relative deprivation in

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general, has been linked to reduced social mobility (Iyer and Pryce 2024). Finally, in view of the overall neighbourhood literature, our results are to some extent in line with G. Galster et al. (2010). They argue that groups of men are 'more influenced by neighbourhood social interactions, social disorder and/or networks or perhaps external forces or stigmatisation' (G. Galster et al. 2010, 2934) than women. They also argue that the means of obtaining employment may consistently be more informal and localised for men in comparison to women.

Lastly, it is important to point out that the operations of the mechanisms inside the SF are a black box: we can demonstrate the observed characteristics of the residents, but we do not have data on the intra-person or intra-household relations that cause this type of network effect. Neither do we have data on initiatives invoked by the government, the Oslo municipality or local communities to relax any negative neighbourhood effects caused by SF. However, in an ongoing paper, we do interviews with residents of selected SF areas in Oslo, which shed additional light on residents' perceptions of SF areas. In relation to this, we emphasise that the SF have been identified using neighbourhood proportions of residents with an immigrant background (first- and second-generation) relative to the total population. However, alternative socio-demographic dimensions may exist that could lead to a different set of identified SF. For example, our identified SF do not distinguish between immigrants of Western or non-Western origin, which has been shown to have consequences for both the neighbourhood selection and the estimated effects (G. C. Galster et al. 2024; Santiago et al. 2024). When we investigate the impacts of the frontiers, we examine heterogeneous effects for immigrants, of whom 81%–85% are of non-Western origin depending on gender. Due to sample size limitations, we leave the analysis of effects for descendants of immigrants as a topic for future research.

As a final note on the SF, we do not explicitly investigate how the effect varies according to the intensity of the treatment, although we do include control variables to ensure that treatment and control groups are comparable with respect to treatment intensity. At the individual-level, we control for the duration of neighbourhood residence. At the neighbourhood-level, we control for the relative length of the SF. A future investigation of varying responses to frontier intensities would statistically benefit from a greater number of neighbourhoods in the study.

7 | Conclusion

We study the impacts on earnings of living in neighbourhoods located close to SF, which exist where there are substantial and sharp divisions in the proportion of immigrants relative to natives residing in spatially adjacent neighbourhoods. We use individual-level longitudinal administrative data and PSM to control for self-selection into neighbourhoods.

The analyses offer unambiguous results regarding the link between SF and immigrant earnings. Living nearby a SF in Oslo does give rise to a negative trend for immigrants, but the effects do not appear to be great in terms of long-term cumulative earned incomes. These results hold irrespective of looking only at SF exposure or looking at the combined effect of SF exposure and neighbourhood residential composition. For native men, the effects are similar to those of immigrant men, but for native women, there is no evidence of any negative effect.

Putting these results into a broader context hints at the role played by SF for labour market integration of immigrants in Norway. Living in proximity to SF influences earnings and their development only to a small extent, suggesting a weak link between SF and economic integration. Immigrant groups tend to have a weaker position than natives on the labour market, which is revealed by higher unemployment rates, lower quality of employment, and, in our data, lower earned income. Nevertheless, our findings show only small effects on earned income for immigrants residing near an SF, suggesting that SF, as an indicator of ethnic residential segregation, do not strongly hamper the earnings potential on average.

It has been argued that SF can potentially hinder the interaction with natives, thus reducing the chances of acquiring language skills and getting access to broader labour market networks, which in turn may dim job and earnings prospects (e.g., Allport 1954). On a more positive note, adjacent neighbourhoods close to SF may be populated by 'bridge-builders', who have more approving attitudes towards the outgroup community (Iyer and Pryce 2024), potentially generating more intergroup interaction and spillover effects beneficial for labour market outcomes. Given our empirical results, we cannot rule out that such mechanisms working in a positive and negative direction have effect and together produce an overall effect of SF that on average appears to be small in the Norwegian capital.

However, even if the effects of SF on earnings are small, from a political point of view, there may still be other arguments for avoiding high residential segregation as well as SF in the urban landscape.

What are the policy implications of our findings? Our conclusion is that immigrants' weaker situation in the Norwegian labour market must be primarily due to circumstances other than the residential segregation that SF give rise to. On a more speculative note, this may instead have to do with individual factors such as lack of language skills, insufficient formal education or relevant professional experience and lack of network contacts leading into the labour market. However, societal factors such as discrimination are also at work hindering successful labour market outcomes.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data is made available for us by Statistics Norway (SSB). They are strictly confidential and cannot be shared. We want to thank Karl-Gunnar Severinsen for assistance in creating the map.

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Endnotes

[1]The registers define a native as a person born in Norway by two Norwegian-born parents. An immigrant is born abroad of two non-native parents (i.e., first generation). When identifying the SF, we also include those born in Norway who have two immigrant parents (i.e., second generation). When we study the effect of SF, we focus on the effect for first-generation immigrants and natives.

[2]By 2018, the percentage of individuals lost due to death or migration is 14.4 for immigrant men, 11.7 for immigrant women, 3.2 for native men and 2.3 for native women.

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